• DRIVEABILITY—NO START/REPEAT STARTER FAILURE—REPEAT STARTER REPLACEMENT • ELECTRICAL—STICKING OR BINDING IGNITION LOCK CYLINDER BETWEEN START AND RUN • ENGINE—STARTING—STARTER STAYING ENGAGED AT START UP	**Article No. 02-19-4**

FORD: 1999-2003 EXCURSION, SUPER DUTY F SERIES

ISSUE

Some 1999-2003 SuperDuty and Excursion vehicles may exhibit the condition of starter staying engaged after releasing key from start position, repeat starter failure or a sticking/binding ignition lock cylinder. This may be due to:

- an ignition lock cylinder binding not fully returning from start to run
- short condition on starter relay circuit 113 (YE/LB) or 1093 (TN/RD)
- sticking starter relay
- an internal short in the Power Distribution Junction Box (PDJB).

ACTION

If one of these conditions occur, inspect the starter and flywheel for damage and replace as necessary. Inspect the starter circuit for shorts including the Power Distribution Junction Box (PDJB) and verify proper cylinder lock function. Service as necessary. Use the following procedure to diagnose the starter circuit and system components.

SERVICE PROCEDURE

To check for short in the starter relay circuit 113:

1. Check terminal #2 at starter relay for voltage with key in run position (move key from “OFF” to “RUN”) with the engine off. If voltage is present, locate and repair short. If no short detected, continue with Step 2.
2. With starter relay terminal #2 removed and vehicle in park, monitor voltage at starter relay terminal after cycling ignition from START to RUN (Do not assist key back to run position). If voltage is present after cycling key returns to Run position, Connect NGS/WDS or Scan Tool and monitor PIDS. Check for KEY_START and KEY_RUN. If the KEY_RUN PID is not seen after any KEY_START PID, the ignition cylinder lock is seizing and must be replaced (See Replacement Applications Heading).
   - For 1999-2001 vehicles, access the GEM (Generic Electronic Module) to view PIDS
   - For 2002-2003 vehicles, access the HEC (Hybrid Electronic Cluster) to view PIDS.

   If key check is functioning properly and voltage is still present, check the starter relay for sticking by swapping with a known good relay and repeat Step 2.

IF ALL OF THE ABOVE CONDITIONS ARE VERIFIED FOR PROPER OPERATION, THE ISSUE MAY BE CAUSED BY AN INTERMITTENT BINDING CONDITION IN THE IGNITION CYLINDER LOCK. PROCEED WITH REPLACEMENT OF KEY-IN-IGNITION WARNING SWITCH (11A127) OR IGNITION CYLINDER LOCK (11582). REFER TO REPLACEMENT APPLICATIONS TO DETERMINE WHICH SERVICE REPLACEMENT PART IS NECESSARY.

Replacement Applications

For vehicles built prior to 12/1/2001 and key-in-ignition warning switch has not been replaced on a prior repair, replace Key-in-Ignition warning switch (XL2Z-11A127-AB). (Refer to the Key in Ignition Warning Switch replacement procedure).

For vehicles built prior to 12/1/2001 and key-in-ignition warning switch has been replaced on a prior repair and vehicle is exhibiting the above conditions, replace the ignition cylinder lock (1L3Z-11582-AA). (Refer to Section 211-04 of the Workshop Manual for repair procedure)

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For vehicles built after 12/1/2001, replace the key-in-ignition warning switch only (XL2Z-11A127-AB). (Refer to the Key in Ignition Warning Switch replacement procedure).

Key In Ignition Warning Switch Replacement Procedure

1. Disconnect the battery ground cable and wait at least one minute.
2. Insert the ignition key and turn it to the RUN position.
3. Insert a punch into the access hole (located in the lower steering column shroud, below the lock cylinder) and press the release button while pulling out the ignition switch lock cylinder.

NOTE
SOME VEHICLES DO NOT HAVE AN ACCESS HOLE AND REQUIRE LOWER SHROUD REMOVAL. REFER TO THE APPROPRIATE MODEL WORKSHOP MANUAL, IGNITION SWITCH LOCK CYLINDER REMOVAL AND INSTALLATION SECTION 211 IF ADDITIONAL DETAIL IS REQUIRED.

4. Remove the Plastic Key chime warning switch by prying upward on the plastic retainer, while sliding the switch away from the key insertion end (Figure1). Refer to the appropriate model Workshop Manual, Key-in-ignition warning switch removal and installation section located in Section 413 if additional detail is required.
5. Install a new Key-In-Ignition warning chime switch (XL2Z-11A127-AB) and reinstall lock cylinder.
6. Verify proper function of the Ignition Switch Lock Cylinder and Key-In- Ignition warning chime switch.

PART NUMBER	PART NAME
XL2Z-11A127-AB	Key-In-Ignition Switch
1L3Z-11582-AA	Ignition Lock Cylinder

OTHER APPLICABLE ARTICLES: NONE
WARRANTY STATUS: INFORMATION ONLY
OASIS CODES: 203000, 203200, 205000, 601300, 603300

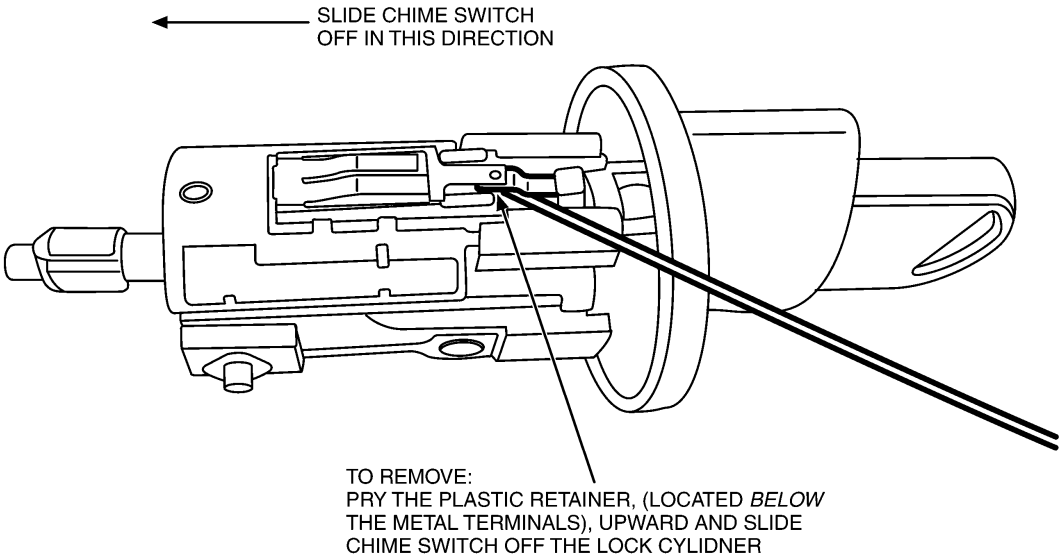

Figure 1 - Article 02-19-4

NOTE: The information in Technical Service Bulletins is intended for use by trained, professional technicians with the knowledge, tools, and equipment to do the job properly and safely. It informs these technicians of conditions that may occur on some vehicles, or provides information that could assist in proper vehicle service. The procedures should not be performed by "do-it-yourselfers". Do not assume that a condition described affects your car or truck. Contact a Ford, Lincoln, or Mercury dealership to determine whether the Bulletin applies to your vehicle.